

Space Physics

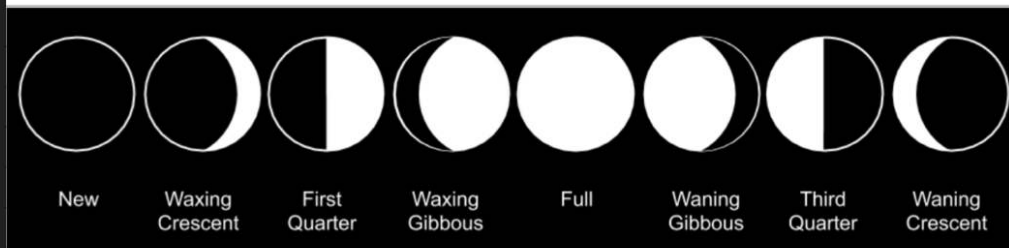
* Earth is a planet that rotates on a tilted axis once in approximately 24 hours. Which explains the apparent daily motion of the Sun and the periodic cycle of day and night.

* Earth orbits the Sun once in approximately 365 days. Which explains the periodic nature of seasons.

When it is summer there are two reasons for higher temperatures:

- 1- Because of the tilt of the axis, the surface of a region is more square on to the Sun, so the Sun's radiation is less spread out
- 2- There are more hours of sunlight.

* It takes approximately one month for the moon to orbit the Earth.



$$\text{Moon's orbital speed} = \frac{2\pi r}{T}$$

r → Radius
T → Period.

The solar system contains:

- One star: The Sun
- Mercury, Venus, Earth, Mars, Asteroid belt Jupiter, Saturn, Uranus, Neptune.
- Dwarf planets: Pluto, asteroids in asteroid belt
- Moons that orbit planets
- Comets and natural satellites.

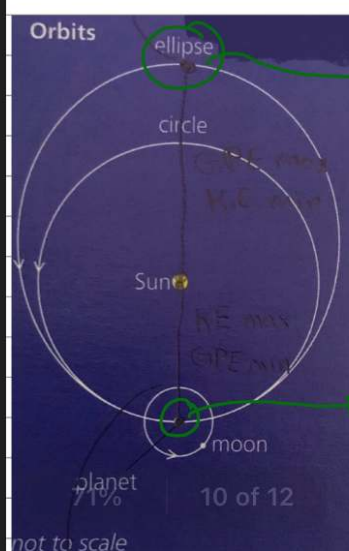
Inner planets

Outer planets

* Planets, minor planets and comets have elliptical orbits. The Sun is not at the center of the orbit except when the orbit is approximately circular.

→ Inner planets are rocky and small.

→ Outer planets are gaseous and large.



→ Minimum gravitational force → Minimum kinetic energy → Maximum GPE

* The strength of the gravitational field

a) at the surface of the planet depends on the mass of the planet.

b) around a planet decreases as the distance from the planet increases.

→ Maximum gravitational force → Maximum kinetic energy → Minimum GPE

- * The Sun contains most of the mass of the smaller system, and this is why the planets orbit the Sun.
- * The force that keeps an object in orbit around the Sun is the gravitational attraction of the Sun.
- * The strength of the Sun's gravitational field decreases and the orbital speeds of the planets decrease as the distance from the Sun increases.
- * An object in an elliptical orbit travels faster when closer to the Sun.

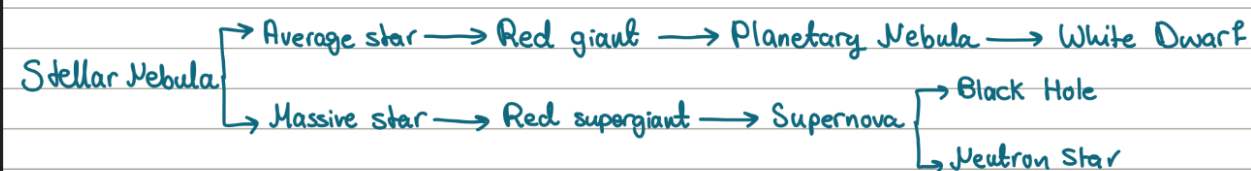
→ The Sun is a star of medium size, consisting mostly of hydrogen and helium. It radiates most of its energy as infrared, visible and ultraviolet regions of the electromagnetic spectrum.

→ The stars are powered by nuclear reactions that release energy. In stable stars, nuclear reactions involve the fusion of hydrogen into helium.

- * Galaxies are each made up of many billions of stars.
 - * The Sun is a star in the galaxy known as the Milky Way.
 - * Other stars that make up the Milky Way are much further away from the Earth than the Sun is from Earth.
 - * Astronomical distances can be measured in light years. Where one light year is the distance travelled in the vacuum of space by light in one year.
- One light year = $9.5 \times 10^{15} \text{ m}$

The life cycle of a star:

- 1- A star is formed from interstellar clouds of gas and dust that contain hydrogen
- 2- A protostar is an interstellar cloud collapsing and increasing in temperature as a result of its internal gravitational attraction.
- 3- A protostar becomes a stable star when the inward force of gravitational attraction is balanced by an outward force due to the high temperature in the center of the star
- 4- All stars eventually run out of hydrogen as fuel for the nuclear reaction.
- 5- Most stars expand to form red giants and more massive stars expand to form red supergiants when most of the hydrogen in the center of the star has been converted to helium.
- 6- A red giant from a less massive star forms a planetary nebula with a white dwarf star at its center.
- 7- A red supergiant explodes as a supernova, forming a nebula containing hydrogen and new heavier elements, leaving behind a neutron star or a black hole at its center.
- 8- The nebula from a supernova may form new stars with orbiting planets



- * The Milky Way is one of many billions of galaxies making up the Universe.
- * The diameter of the Milky Way is approximately 100000 light years.

→ Red shift: An increase in the observed wavelength of electromagnetic radiation emitted from receding stars and galaxies.

→ The light emitted from distant galaxies appears redshifted in comparison with light emitted on Earth.

- * The redshift in the light from distant galaxies is evidence that the Universe is expanding and supports the Big Bang theory.

- * Cosmic microwave background radiation (CMBR): The microwave radiation of a specific frequency is observed at all points in space around us
- * The CMBR was produced shortly after the Universe was formed and that this radiation has been expanded into the microwave region of the electromagnetic spectrum as the universe expanded
- * The speed (v) at which a galaxy is moving away from Earth can be found from the change in wavelength of the galaxy's starlight due to the redshift.
- * The distance (d) of a far galaxy can be determined using the brightness of a supernova in that galaxy.

Hubble constant: H_0 : The ratio of the speed at which the galaxy is moving away from Earth to its distance from Earth

$$H_0 = \frac{v}{d}$$

- * The current estimate for H_0 is 2.2×10^{-18} per second.

→ The equation: $\frac{d}{v} = \frac{1}{H_0}$ represents an estimate for the age of the universe and that this is evidence for the idea that all the matter in the Universe was present at a single point.